A Spatial Alignment Method for UAV LiDAR Strip Adjustment in Nonurban Scenes

Yanfeng Gu, *Senior Member, IEEE*, Zhen Xiao, and Xian Li, *Member, IEEE*

***Abstract*— Light detecting and ranging (LiDAR) strip adjustment is a key prerequisite for subsequent applications based on point cloud data since it inevitably suffers from spatial discrepancies caused by laser ranging errors, mounting errors, etc. Most current LiDAR strip adjustment methods rely on the extraction of structural features, which are often unsuitable for nonurban scenes. Alternative strip adjustment methods based on correspondence distance minimalization ignore spatial alignment. To overcome these limitations, this article presents an accurate spatial alignment method for an unmanned aerial vehicle (UAV) LiDAR strip adjustment in nonurban scenes. First, we construct a novel point cloud feature descriptor called Spherical Shell Point Feature (SSPF) to extract multidimensional nonstructural features that are robust to nonurban point clouds. The constructed SSPF is then combined with point coordinates to generate embedded features, which simultaneously consider the point coordinates and spatial alignment. Finally, the embedded features are used by a two-stage matching method to match pair-wise points of two adjacent strips. The proposed method is validated on two nonurban datasets collected by two types of LiDARs, which reduces the digital surface model (DSM) discrepancies by 0.252 and 0.221 m, respectively, and proves its superiority compared to mainstream strip adjustment methods as well.**

***Index Terms*— Light detection and ranging (LiDAR), nonurban point cloud, point cloud descriptor, strip adjustment.**

I. Introduction

RECENTLY, the light detecting and ranging (LiDAR) system has developed rapidly for its ability to quickly and accurately acquire three-dimensional (3-D) spatial information, playing an increasingly important role in classification [1], building reconstruction [2], forest index estimation [3] and other research fields [4], [5], [6]. LiDAR is an integrated system, including a laser scanner, a global positioning system (GPS), and an inertial measurement unit (IMU) [7], and the latter two compose the position and orientation system (POS) for real-time measurement of LiDAR position and attitude information. LiDAR mounted on an unmanned aerial vehicle (UAV) platform offers the advantages of low cost, high point density, and convenient data acquisition, which has attracted much attention from researchers [8].

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The authors are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: xianli@hit.edu.cn).

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UAV LiDAR data acquisition is usually carried out in a strip-wise pattern [9], and multiple strips are often required to cover the whole observed scene due to a limited field of view. The points of the same ground objects in adjacent strips, however, typically provide different 3-D spatial coordinates (spatial discrepancies), which is caused by laser ranging errors, mirror angle scale errors, mounting errors, POS systematic errors, etc. [10]. The spatial discrepancies are virtually inevitable in real applications and result in significantly deviated results if exploiting the original LiDAR data directly. To reduce the spatial discrepancies, a variety of strip adjustment methods have been proposed, which can be classified into sensor system-driven methods and data-driven methods.

The main idea of sensor system-driven methods is to employ LiDAR geolocation equations as a strip adjustment model to reduce mounting errors [11]. These methods require the original observations of GPS/IMU and laser scanner or at least the trajectory and the point clouds with time stamps [11]. The sensor system-driven methods carry out strip adjustment by tie elements, such as planes, lines, and points between different strips or between one strip and ground control points. The planes or surfaces are easily extracted in urban areas, which are employed in [12] and [13] as tie elements. For example, Filin and Sagi [12] used least squares with natural and artificial surfaces to estimate the boresight angle and laser offset of the airborne LiDAR system. Subsequently, Skaloud and Lichti [13] solved the boresight calibration problem of the sensor system by extracting and matching planar features. The linear features were also employed [14], [15], particularly in man-made environments. In [14], Chan et al. proposed a boresight self-calibration method based on a combination of the planar features and catenary features. Ravi et al. [15] presented a LiDAR system calibration procedure to directly estimate the mounting parameters based on conjugate linear features in overlapping point clouds. Instead of using a planer or linear features, Kumari et al. [11] proposed a method to calculate the boresight angles of the airborne LiDAR system based on point-plane distance minimization. Although the aforementioned methods reduce the spatial discrepancies between strips, their model parameters are not once and for all. In addition, the raw observations of laser and POS are often unavailable.

The data-driven strip adjustment methods aim to minimize the corresponding distances between the overlapping region of different strips via estimating a rotation matrix and a translation vector. They merely require point clouds

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without involving any observations of GPS, IMU, and laser scanner. As pioneers, Besl and Mckay [16] proposed a classical iterative closest point (ICP) algorithm to solve the 3-D-shape data registration problem. Subsequently, a variety of improvements were successfully developed based on ICP, including point-point [16], [17], [18], [19], [20], [21], point-plane [22], [23], [24], and plane-plane [25]. These ICP-based methods are applied to various data processing, including LiDAR strip adjustment [26]. For instance, Glira et al. [26] developed an automatic strip adjustment method based on ICP, which directly established point correspondences iteratively in overlapping area.

In addition to the aforementioned methods, there are also some methods based on structural feature extraction [27], [28], [29], [30], [31], [32], [33], [34]. Instead of using iterative optimization, these methods directly calculate point cloud transformation by finding the correspondence among the extracted features from two adjacent strips. Among them, several methods [27], [28], [29], [30] extracted the edge line features of buildings since they are more reliable and suitable for strip adjustment. For example, Lee et al. [27] and Habib et al. [28] proposed to segment the building patches for the extraction of ridgelines, and used the similarity of these ridgeline segments to adjust strips. Similarly, the intersection of the T-shaped roof ridges was used to adjust urban strips and achieved good results [29], but it was non-effective without T-shaped buildings in strips. In addition, Yang et al. [30] presented match lines, which were extracted from artificial buildings and pole-like objects by a combination of semantic and geometric features. Alternative methods perform strip adjustment by extracting planes or surfaces, which are common in urban scenes, and minimizing the distances between them. Filin [31] researched the influence of surface characteristics on strip adjustment, and experimental results revealed that sloped surfaces with different orientations can improve the strip adjustment results. Subsequently, a general method for determining discrepancies automatically based on segmentation of the overlap and surface feature extraction was proposed [32]. Liu et al. [33] presented a strip adjustment method using planar features that were extracted by using the smallest Hausdorff distance. Also using plane features, Song et al. [34] employed the RANSAC algorithm to extract the building plane and used the minimum Hausdorff distance to determine the line pair.

Instead of extracting structural features, few methods adopted handcrafted features [35], [36], [37], [9], [38], [39]. For example, Rusu et al. [35] and [36] extracted local multidimensional features of points, called persistent feature histograms (PFH) and fast point feature histograms (FPFH), by establishing a new local reference frame (LRF) and counting the histogram of the angles and distances between the neighbor points. Weber et al. [37] presented an automatic registration method by employing FPFH in coarse registration. In [9] and [38], conjugate features were adopted for strip adjustment, which requires high-resolution aerial images of the same observed area.

Although earlier reported methods provide favorable results in LiDAR strip adjustment, there are still several limitations. One major limitation is that they rely largely on structural elements such as artificial objects, which are rarely found in nonurban scenes in practice. Nonurban scenes usually consist of terrain and/or forest scenes as the main background but rarely artificial objects [40], and thus their spatial features typically have no obvious structural characteristics. Another limitation is that the methods based on correspondence distance minimalization always ignore the spatial alignment, which makes these methods suffer from the sparse and uneven distribution of point clouds.

Motivated by these existing limitations, we present an accurate spatial alignment method for UAV LiDAR strip adjustment in nonurban scenes. In this method, we first construct a novel point cloud feature descriptor called Spherical Shell Point Feature (SSPF), which is for extracting multidimensional features for nonurban point clouds. Compared with the existing point feature descriptors, SSPF is independent of structural elements (which rarely occur in nonurban areas) and LRF construction (which is unreliable due to cluttered ground objects). Subsequently, we combine SSPF with point coordinates to generate embedded features, which take both the point coordinates and spatial alignment into account. Finally, the embedded features are used by a two-stage matching method to elegantly match pair-wise points of two adjacent strips. Our method proves to be effective on two challenging datasets collected by different UAV LiDAR systems. The major contributions are summarized as follows.

1) We propose an accurate spatial alignment method to solve the UAV LiDAR spatial alignment problem in strip adjustment of nonurban scenes, which takes both global correspondence distance minimization and spatial alignment into account without any extraction of structural elements.
2) We introduce a novel point feature descriptor called "SSPF," which gains high descriptiveness by extracting multidimensional features from multiscale neighborhood information. Its key advantages, including rotation invariance, insensitivity to point cloud density and distribution, and robustness to noise that are important for point cloud transformation and strip adjustment.
3) A two-stage matching method is designed for pair-wise point matching, which makes use of point coordinates and SSPF, improving the accuracy of correspondences.

The rest of this article is organized as follows. Section II describes the proposed strip adjustment method in detail. A thorough experimental evaluation and discussion of the results are given in Section III. Section IV outlines the conclusion and potential future research directions.

II. Proposed Method

A. Overall Framework

We propose an accurate spatial alignment method for UAV LiDAR strip adjustment in nonurban scenes. Fig. 1 shows its overall framework. The inputs are two original adjacent strips sharing a certain overlapping area in nonurban scene, which are performed the outlier removal [37] and overlap identification operations to improve their quality. To enhance

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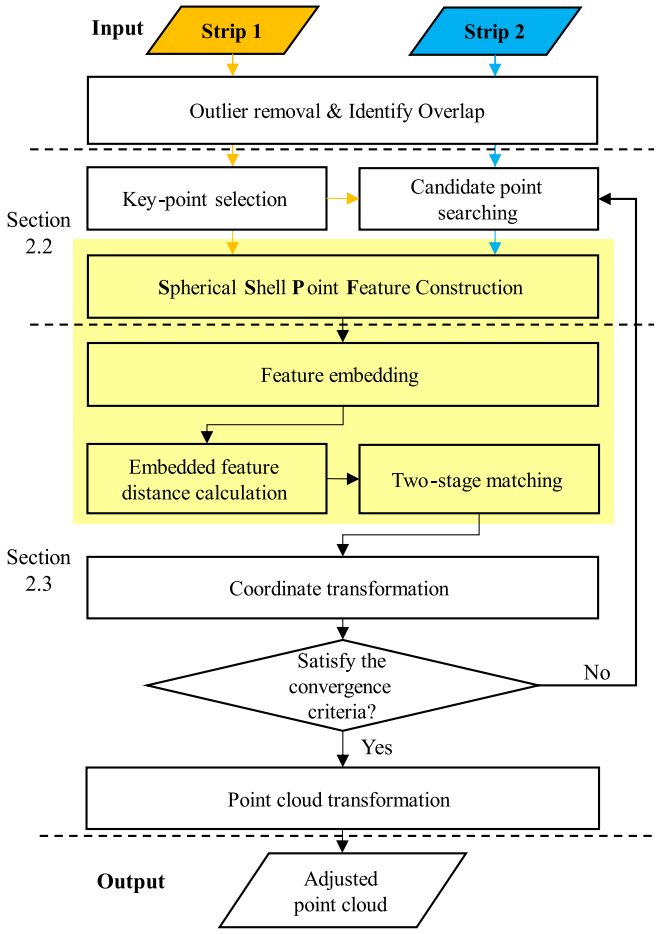


Fig. 1. Overall framework of the proposed strip adjustment method.

computational efficiency, key-point selection and candidate point searching are carried out. These points are used to extract multidimensional nonstructural features with the proposed point cloud feature descriptor, called SSPF. Taking both the point coordinates and spatial alignment into account, we then combine SSPF with point coordinates to generate embedded features. To match pair-wise points of two adjacent strips, a two-stage matching method are designed using the embedded features. Through the coordinate transformation in several iterations, the adjusted point cloud is outputted via point cloud transformation, which contains all points from the inputs in a common global coordinate system. For the convenience of representation, the first strip is denoted as strip1, the point cloud in strip1 is P_1, and the second strip is denoted as strip2 and P_2 in this article.

B. SSPF Construction

According to the existing point feature descriptor, a unique and robust LRF or a stable normal is important for both robustness and descriptiveness. Unlike the buildings and smooth ground surfaces in urban scenes, the scattered vegetation and undulating terrain in nonurban scenes create much uncertainty, making the construction of LRF or the estimation of a normal to be unreliable. Instead of LRF construction and normal estimation, we here propose an SSPF descriptor to extract multidimensional features for spatial alignment in UAV LiDAR strip adjustment. The main idea of SSPF is to divide the neighborhoods of the points into multiscale spherical shell shapes and then extract multidimensional features within these spherical shells at different scales.

1) SSPF Construction: Let p_0 be a point in the point cloud P. Its neighborhood points are divided into several spherical shells centered at p_0 according to a set of increasing radiuses. The point set $\mathbf{ps}_m \in \mathbb{R}^{N_m \times 3}$ in the mth spherical shell can be expressed as

$$\mathbf{ps}_m = \left\{ p \middle| \, R_m < r \leq R_{m+1},\ m = 1, \ldots, M \right\} \tag{1}$$

where N_m is the number of points in $\mathbf{ps}_m$, r is the Euclidean distance between p_0 and p, R_m and R_{m+1} are the inner and outer radius of the mth spherical shell, respectively. We suppose $R_{m+1} = R_m + t$, and t denotes the thickness of each spherical shell. M is the number of spherical shells.

For each spherical shell, we first construct two parallel planes satisfying two constraints to determine them uniquely: 1) all points in this spherical shell are sandwiched between the two planes, and 2) the distance between the two parallel planes is minimized. In this way, three distances can be calculated: the distance between the two planes (i.e., d_m^1), and the distances from p_0 to the two planes (i.e., d_m^2 and d_m^3). Fig. 2(a) shows the schematic of the two plane's construction. We then concatenate the three distances into a feature vector $\mathbf{d}_m \in \mathbb{R}^{1\times 3} = [d_m^1, d_m^2, d_m^3]$. Considering that the elements in $\mathbf{d}_m$ are interdependent each other in both two cases as shown in Fig. 2(b) and (c), we rewrite $\mathbf{d}_m \in \mathbb{R}^{1\times 2} = [d_m^1, d_m^{23}]$, where d_m^{23} is the maximum value between d_m^2 and d_m^3.

We extend the above operations into a multiscale representation (i.e., M), which generates SSPFs $\mathbf{f}_1 \in \mathbb{R}^{2M\times 1}$ as following:

$$\mathbf{f}_1 = [\mathbf{d}^1, \mathbf{d}^{23}]^T \tag{2}$$

where $(\cdot)^T$ is the transpose operator, $\mathbf{d}^1 \in \mathbb{R}^{M\times 1} = [d_1^1, \ldots, d_M^1]$ represents the spatial information of the points within the spherical shell, and $\mathbf{d}^{23} \in \mathbb{R}^{M\times 1} = [d_1^{23}, \ldots, d_M^{23}]$ represents the relation between p_0 and its neighborhood. In realistic scenes, the environment around the same ground object in different strips is consistent; thus, the SSPFs of two points (which represent the same object from different strips) are similar. Conversely, the SSPFs are most likely not similar when the points represent different objects.

The calculation of the distance can be represented by

$$\begin{aligned} &\min_{\substack{\mathbf{w}\in\mathbb{R}^{3\times 1} \\ D\in\mathbb{R}^1}} \left\| \mathbf{ps}_m \cdot \mathbf{w} + D \cdot \mathbf{1}_{N_m\times 1} \right\|_\infty \\ &\text{s.t.} \quad \|\mathbf{w}\|_2 = 1 \end{aligned} \tag{3}$$

where $\mathbf{1}_{N_m\times 1} = [1, 1, \ldots, 1]^T \in \mathbb{R}^{N_m\times 1}$, $\|\cdot\|_\infty$ is ℓ_∞-norm and $\|\cdot\|_2$ denotes ℓ_2-norm. $\mathbf{w} \in \mathbb{R}^{3\times 1}$ and $D \in \mathbb{R}^1$ represents the normal direction of the plane and the intercept of the projection in the normal direction, respectively. Once we obtain the $\mathbf{w}$ and D, $[d_m^1, d_m^{23}] = [2\max\{\mathbf{ps}_m \cdot \mathbf{w} + D \cdot \mathbf{1}_{N_m\times 1}\}, \max\{|(\mathbf{p}_0 - \mathbf{ps}_m) \cdot \mathbf{w}|\}]$ can be calculated, where $|\cdot|$ is taking the absolute value.

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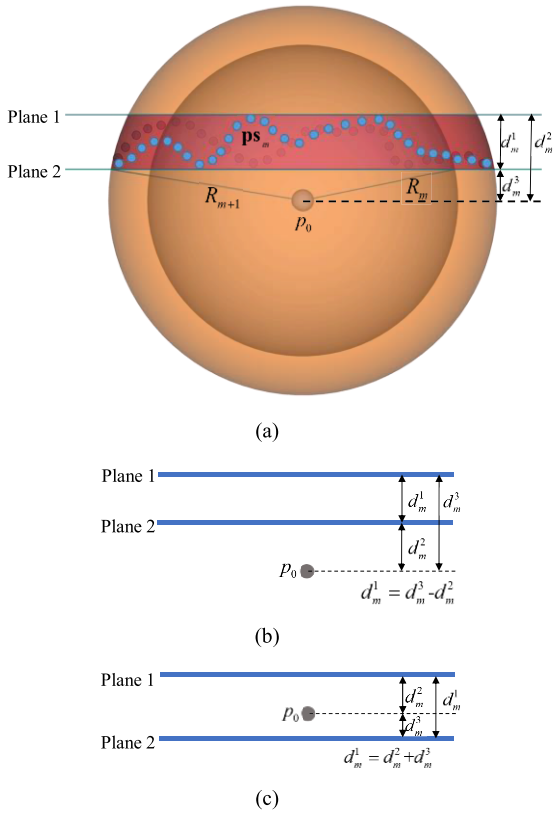


Fig. 2. Schematic of SSPF for the mth spherical shell. (a) SSPF schematic. (b) First case. (c) Second case.

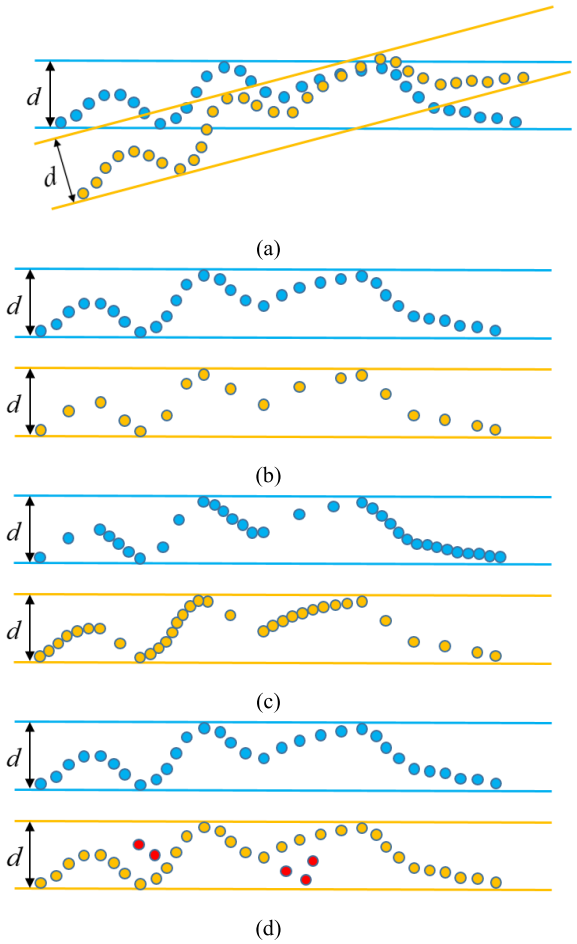


Fig. 3. Illustrations of SSPF properties. The blue and orange points represent different point sets, and the red points are noise points. (a) Rotation. (b) Down-sampling. (c) Point distribution. (d) Noise.

2) SSPF Properties: We construct SSPF since it has four key properties, which are important to match pair-wise points as discussed below:

a) Rotation invariance: Thanks to the division strategy of multiscale spherical shell and the calculation method of $\mathbf{d}^1$ and $\mathbf{d}^{23}$, SSPF remains unchanged regardless of the rotation of the point cloud around any axis by any angle, as shown in Fig. 3(a). Apply a rotation matrix $\mathbf{R}_x \in \mathbb{R}^{3\times3}$ to the subset $\mathbf{ps}_m$, (3) can be rewritten as

$$\begin{aligned}\left\|\mathbf{ps}_m \cdot \mathbf{w} + D \cdot \mathbf{1}_{N_m\times1}\right\|_\infty &= \left\|\mathbf{ps}_m \cdot \mathbf{R}_x \cdot \mathbf{R}_x^{-1} \cdot \mathbf{w} + D \cdot \mathbf{1}_{N_m\times1}\right\|_\infty \\ &= \left\|\mathbf{ps}'_m \mathbf{w}_1 + D \cdot \mathbf{1}_{N_m\times1}\right\|_\infty \end{aligned} \tag{4}$$

where $\mathbf{ps}'_m \in \mathbb{R}^{N_m\times3} = \mathbf{ps}_m \cdot \mathbf{R}_x$ is the point set after rotation, and $\mathbf{w}_1 \in \mathbb{R}^{3\times1} = \mathbf{R}_x^{-1} \cdot \mathbf{w}$ represents the new normal direction of the plane for $\mathbf{ps}'_m$. Then plug $\mathbf{ps}'_m$, $\mathbf{w}_1$ and D into the calculation of $\mathbf{d}^1$ and $\mathbf{d}^{23}$, which are found to remain constant. This property is important for point cloud transformation.

b) Permissibility of down-sampling: SSPF only depends on the points located on the profile of each subset in a spherical shell. This allows a reasonable down-sampling that keeps the profile unchanged. A uniform down-sampling of $\mathbf{ps}_m$ can be expressed as $\Lambda \cdot \mathbf{ps}_m = [\,\mathbf{0}_{\tilde{N}_m\times(N_m-\tilde{N}_m)}\ \mathbf{I}_{\tilde{N}_m\times\tilde{N}_m}\,] \cdot [\,\mathbf{ps}_{m,1}\ \mathbf{ps}_{m,2}\,]^T$, where $\Lambda \in \mathbb{Z}^{\tilde{N}_m\times N_m}$ is the sample matrix, $\mathbf{ps}_{m,1} \in \mathbb{R}^{(N_m-\tilde{N}_m)\times3}$ and $\mathbf{ps}_{m,2} \in \mathbb{R}^{\tilde{N}_m\times3}$ are abandoned and sampled points, respectively, $\tilde{N}_m \in \mathbb{R}$ refers to the number of points in $\mathbf{ps}_{m,2}$. The distribution of $\mathbf{ps}_{m,2}$ and $\mathbf{ps}_m$ is approximately identical, as well as $\mathbf{ps}_{m,2} \cdot \mathbf{w}$ and $\mathbf{ps}_m \cdot \mathbf{w}$. Equation (3) can be rewritten as

$$\begin{aligned}\left\|\mathbf{ps}_m \cdot \mathbf{w} + D \cdot \mathbf{1}_{N_m\times1}\right\|_\infty &\approx \left\|\mathbf{ps}_{m,2} \cdot \mathbf{w} + D \cdot \mathbf{1}_{\tilde{N}_m\times1}\right\|_\infty \\ &= \left\|\Lambda \cdot \mathbf{ps}_m \cdot \mathbf{w} + D \cdot \Lambda \cdot \mathbf{1}_{N_m\times1}\right\|_\infty. \end{aligned} \tag{5}$$

Equation (5) means that the $\mathbf{w}$ and D calculated by $\mathbf{ps}_{m,2}$ are rarely changed compared to the results by $\mathbf{ps}_m$. Fig. 3(b) gives an example that there is no significant change in the SSPF calculation before and after the down-sampling processing of the point clouds. This property can greatly reduce the computation complexity for strip adjustment, especially for large-scale LiDAR data.

c) Insensitivity to point distribution: Different flight directions and diverse scanning modes lead to different point cloud distributions on the same object in separate strips. SSPF is insensitive to uneven and irregular point distribution caused by various and unavoidable reasons. For the distribution of random nonuniform down-sampling, the analysis of (5) is still applicable. This property allows SSPF to carry out more steadily, as shown in Fig. 3(c).

d) Robustness to noise: SSPF is robust to noise because there is no impact if the noise points are located inside the two parallel planes. For the point cloud containing some noise

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points, (3) can be rewritten as

$$\begin{aligned}&\left\|\begin{bmatrix}\mathbf{ps}_m\\ \mathbf{ps}_n\end{bmatrix}\cdot\mathbf{w}+D\cdot\begin{bmatrix}\mathbf{1}_{N_m\times 1}\\ \mathbf{1}_{N_n\times 1}\end{bmatrix}\right\|_\infty\\&=\left\|\begin{bmatrix}\mathbf{ps}_m\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_m\times 1}\\ \mathbf{ps}_n\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_n\times 1}\end{bmatrix}\right\|_\infty\\&=\max\left\{\left\|\mathbf{ps}_m\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_m\times 1}\right\|_\infty,\left\|\mathbf{ps}_n\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_n\times 1}\right\|_\infty\right\}\end{aligned}\tag{6}$$

where $\mathbf{ps}_n\in\mathbb{R}^{N_n\times 3}$ represents the noise points, $N_n\in\mathbb{R}$ is the number of the noise points. As for the noise points attached to the interior of $\mathbf{ps}_m$ (considering the outlier removal), we can find that $\|\mathbf{ps}_m\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_m\times 1}\|_\infty > \|\mathbf{ps}_n\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_n\times 1}\|_\infty$. In this case, (6) is identical to $\|\mathbf{ps}_m\cdot\mathbf{w}+D\cdot\mathbf{1}_{N_m\times 1}\|_\infty$, which means the noise points rarely impinge the calculation in (3). In addition, SSPF is multidimensional, which further improves its robustness. This property is illustrated in Fig. 3(d).

For large-scale UAV LiDAR data, it is unfeasible to select each point (up to several million points). Thus, a small number (hundreds) of key points are selected in the overlapping area. Considering the properties of SSPF, we employ a uniform sampling method [26] to make the key points follow an even distribution throughout the overlapping area. Then the candidate points, which will become corresponding to the key points, are searched in P_2 within a fixed radius centered on each key point. Only the key points and the candidate points need to calculate SSPF.

C. Pair-Wise Points Matching

ICP is typically conducted by a one-to-one correspondence of points in different point clouds, so the accuracy of corresponding points determines the quality of the registration results. Current ICP-based methods merely employ the point coordinates but ignore spatial alignment, leading to poor strip adjustment results in nonurban scenes. We design a two-stage matching method that employs both ICP and the embedded features to ensure accurate correspondences.

1) Embedded Feature Generation: Although the point coordinates of the same ground object are inconsistent in different strips, they are still relatively closer than those unrelated points. This is the basic principle of ICP-based methods. Therefore, the point coordinates are not abandoned but combined with SSPF to generate embedded features. The point coordinates x, y, z consist of $\mathbf{f}_2\in\mathbb{R}^{3\times 1}=[x,y,z]^T$. We then stack SSPF $\mathbf{f}_1$ and point coordinates $\mathbf{f}_2$ to generate the embedded features as

$$\mathbf{f}=[\mathbf{f}_1^T,\mathbf{f}_2^T]^T\tag{7}$$

where $\mathbf{f}\in\mathbb{R}^{(2M+3)\times 1}$ represents the embedded features, which will be employed to match pair-wise points next.

2) Embedded Feature Distance Calculation: Before pair-wise point matching, the distances $\Delta\mathbf{f}^n\in\mathbb{R}^{2\times 1}$ between the embedded features of points are calculated as follows:

$$\Delta\mathbf{f}^n=f\left(\mathbf{f}^n,\mathbf{f}^0\right)=[\Delta\mathbf{f}_1^n,\Delta\mathbf{f}_2^n]^T,\quad n=1,2,\ldots,N\tag{8}$$

where $f(\cdot)$ represents the distance calculation function, $\mathbf{f}^0$ denotes the embedded features of the key point in P_1, and $\mathbf{f}^n$ is calculated by the nth candidate point in P_2. N is the number of candidate points. For convenience, $\mathbf{d}^1$ in (2) is denoted as $\mathbf{a}$ and $\mathbf{d}^{23}$ as $\mathbf{b}$, then $\Delta\mathbf{f}_1^n$ is calculated as follows:

$$[\Delta\mathbf{a}_n]_m=|[\mathbf{a}_n]_m-[\mathbf{a}_0]_m|,\quad [\Delta\mathbf{b}_n]_m=|[\mathbf{b}_n]_m-[\mathbf{b}_0]_m|\tag{9}$$

where $\Delta\mathbf{a}_n,\Delta\mathbf{b}_n\in\mathbb{R}^{1\times M}$ represent the distance difference between two points according to their multiscale spherical shells, $[\cdot]_m$ donates the mth element of a vector.

To improve the adaptability of SSPF, instead of fixing a threshold, the binarized distances between $\mathbf{f}_1$ are adopted, which is better applicable to different data. Equation (9) can be rewritten as follows:

$$\begin{aligned}[\Delta\mathbf{a}_n]_m&=\frac{1}{2}\left(\operatorname{sign}\left([\Delta\mathbf{a}_n]_m-\operatorname{mid}\left(\mathbf{e}_{M,m}\mathbf{A}\right)\right)+1\right)\\[\Delta\mathbf{b}_n]_m&=\frac{1}{2}\left(\operatorname{sign}\left([\Delta\mathbf{b}_n]_m-\operatorname{mid}\left(\mathbf{e}_{M,m}\mathbf{B}\right)\right)+1\right)\end{aligned}\tag{10}$$

where sign means taking positive and negative signs, mid represents the operation of taking the median, $\mathbf{e}_{M,m}\in\mathbb{R}^{M\times 1}$ is for picking a row from a matrix, the mth element in $\mathbf{e}_{M,m}$ equals to 1 and other elements equal to 0, and $\mathbf{A}\in\mathbb{R}^{M\times N}=[\Delta\mathbf{a}_1^T,\ldots,\Delta\mathbf{a}_N^T]$, $\mathbf{B}\in\mathbb{R}^{M\times N}=[\Delta\mathbf{b}_1^T,\ldots,\Delta\mathbf{b}_N^T]$ are generated by stacking multi vectors in (9). We obtain $\Delta\mathbf{f}_1^n$ of all the N candidate points by

$$\Delta\mathbf{f}_1^n=\sum\Delta\mathbf{a}_n+\Delta\mathbf{b}_n.\tag{11}$$

As for $\Delta\mathbf{f}_2^n$, it is expressed as $\Delta\mathbf{f}_2^n=\left\|\mathbf{f}_2^n-\mathbf{f}_2^0\right\|_2$. Finally, the distances between the embedded features of two points are obtained as (8).

3) Two-Stage Matching: For acceleration of convergence, we design a two-stage matching method, which is achieved by assigning different distance weightings in the two stages. The first stage is mainly dominated by the point coordinates, with the distance $\Delta\mathbf{f}^n$ between the embedded features gaining weight, in which the first term is smaller than the second term. The main idea of this stage is that the embedded features with this weighting reduce the substantial spatial discrepancies between different point clouds rapidly, providing fewer but more possible candidate points for pair-wise point matching. In this way, the probability of correct point pairing is improved and the computation burden is reduced. The second stage is mainly dominated by the SSPF descriptor, with the distance $\Delta\mathbf{F}^n$ between the embedded features gaining opposite weighting, in which the second term is smaller than the first term. The significance of this stage lies in further reducing the residuals in the first stage by spatial alignment for better strip adjustment results.

The two-stage matching is implemented as follows. After calculating the distance between embedded features, the criteria for corresponding points pairing are as follows:

$$D_n^{(i)}=\omega^{(i)}\cdot\Delta\mathbf{f}^n\tag{12}$$

where $D_n^{(i)}\in\mathbb{R}$ is the weighted distance between the key point and the nth candidate point, and $\omega^{(i)}\in\mathbb{R}^{1\times 2}=[\omega_1^{(i)},\omega_2^{(i)}]$ represents the weighting vector, where $\omega_1^{(i)}>0$ corresponding to the weight of $\Delta\mathbf{f}_1^n$ and $\omega_2^{(i)}>0$ is the weight of $\Delta\mathbf{f}_2^n$. The i represents the number of iterations. For convenience,

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TABLE I
DATASET INFORMATION

Dataset		Coverage area(m)	Points	Average density(pts/m^2)	Overlapping ratio
Inner Mongolia	1st strip	197 × 367	17,853,500	247	80%
	2nd strip	195×402	19,086,092	243	
Yellow River	1st strip	326×355	2,472,059	21	90%
	2nd strip	318×369	2,475,383	21	

let $\omega_2^{(i)} = 1$, and ω_2^i are determined by (13)

$$\omega_1^{(i)} = \begin{cases} 0, & \Delta e_2^{(i-1)} \geq \varepsilon_1 \\ \dfrac{2e_2^{(i-1)}}{e_1^{(i-1)}}, & \Delta e_2^{(i-1)} < \varepsilon_1 \end{cases} \tag{13}$$

where $e_1^{(i)} \in \mathbb{R}$ and $e_2^{(i)} \in \mathbb{R}$ are the residuals of $\Delta \mathbf{f}_1$ and $\Delta \mathbf{f}_2$ in the ith iteration, respectively. $\varepsilon_1 \in \mathbb{R}$ is the threshold for changing the stage, and $\Delta e_2^{(i)} = e_2^{(i)} - e_2^{(i-1)}$.

In each iteration of ICP, the two points with minimum weighted distance $D_n^{(i)}$ are regarded as corresponding points, the corresponding points are used to calculate a rotation matrix $\mathbf{R}^{(i)} \in \mathbb{R}^{3\times3}$ and a translation vector $\mathbf{t}^{(i)} \in \mathbb{R}^{3\times1}$. Note that the point correspondence matching in each iteration is based on embedded features, while the transformation computation is based on point coordinates. The convergence criterion to judge whether the strip adjustment is successful can be expressed as follows:

$$\frac{\sum D^{(i)} - \sum D^{(i-1)}}{\sum D^{(i)}} \leq \varepsilon_2 \tag{14}$$

where $\varepsilon_2 \in \mathbb{R}$ is the threshold for stopping the iteration. If it is not met, the process restarts from the two-stage matching with new candidate points. After the pair-wise point matching, the final rotation matrix $\mathbf{R} \in \mathbb{R}^{3\times3}$ and translation vector $\mathbf{t} \in \mathbb{R}^{3\times1}$ can be calculated as

$$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} = \prod \begin{bmatrix} \mathbf{R}^{(i)} & \mathbf{t}^{(i)} \\ \mathbf{0} & 1 \end{bmatrix}. \tag{15}$$

We then apply the rotation matrix $\mathbf{R}$ and translation vector $\mathbf{t}$ to P_2. Finally, one single point cloud containing points of different strips can be obtained.

III. EXPERIMENT RESULTS AND ANALYSIS

We perform experiments on two challenging nonurban datasets: Inner Mongolia and Yellow River, to evaluate the effectiveness of the proposed method. Two objective metrics: Euclidean distance and digital surface model (DSM) discrepancies are used to evaluate the quality of strip adjustment. 1) Euclidean distance: select a subset from the P_1 in the overlapping area. For the points in the subset, the corresponding points with the minimum Euclidean distance are found in the P_2. Calculate the mean and root mean square error (RMSE) of the Euclidean distance to evaluate the results. 2) DSM discrepancies: generate DSMs of two point clouds and calculate the discrepancies between them, including mean and RMSE.

A. Data Description and Experimental Setup

1) Dataset Description: The Inner Mongolia dataset was collected by Velodyne's Puck Lite with Applanix APX-15 GNSS system in the Hunshadake Sandy Land of Inner Mongolia, China, in 2021. It mainly consists of grassland sand dunes, in which there were more herbs, followed by shrubs and fewer trees. The LiDAR is a 16-line scanner with a horizontal angle of 360° and a vertical angle of 30°. The flying height is 80 m with a speed of 5 m/s. The Yellow River dataset was collected by RIEGL's UAV-MINI1 with APX15 in Yellow River Wetland Park, China, in 2020. The collection area is a wetland environment with vegetation, including Suaeda, reed, and Spartina alterniflora. The LiDAR is a single line with a horizontal field of view of 360°. The flying height is 80 m with a speed of 5 m/s. Fig. 4 shows the two datasets with regional coverage and overlap. More details can be found in Table I.

2) Parameter Settings: To improve computational efficiency, we carried out a uniform down-sampling on the original datasets. The distance threshold between two points is set as 0.2 m. The parameters of our method are set as follows: The maximum range of the neighborhood $r_{\max}$ is set as 10 m. A larger range means that more spatial features can be extracted, but it also means more computation and more likely feature redundancy. The shell thickness t is set as 1 m, which is guaranteed to represent spatial features without hiding details. The radius of the search area for candidate point searching is set as 2 m considering the first stage of two-stage matching. More candidate points are for matching with a larger search radius, which means more computation. The threshold ε_1 is set as 0.1 and ε_1 is set as 0.01. Note that the proposed method has the same parameters in both two datasets.

B. Experimental Results and Analysis

1) Quantitative Evaluation and Analysis: To analyze the performance of the proposed method, point-to-point ICP [16], point-to-plane ICP [26], and FPFH [37], [36] are selected as the mainstream methods for performance comparison. In point-to-plane ICP, the radius for plane construction is set as 1 m, which is mainly due to the characteristics of the irregular objects in the data. In FPFH, the radius of the neighborhood for point descriptor calculation is set as 2 m, and the points within 2 m radius are used to construct LRF.

Tables II and III report the quantitative evaluation of the two datasets. In the Inner Mongolia dataset, the mean of Euclidean distance is reduced from 0.219 to 0.081 m, and the mean of DSM discrepancies is reduced from 0.412 to 0.164 m. The mean and RMSE of both Euclidean distance

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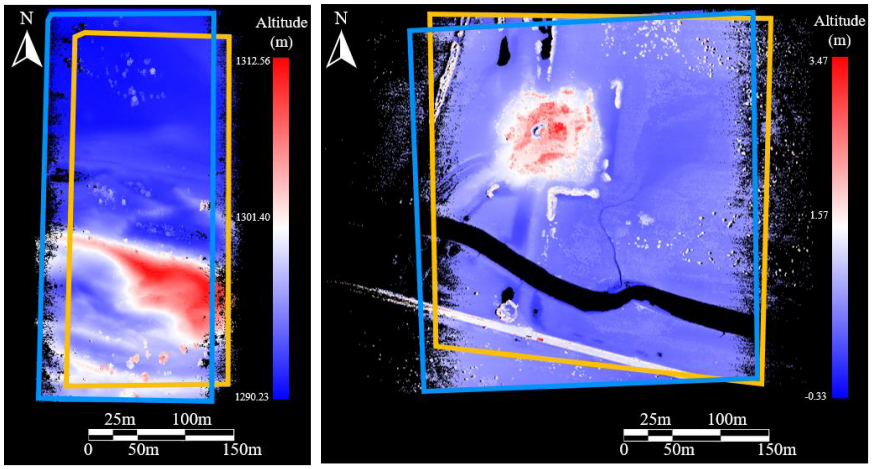


Fig. 4. Two datasets, the orange box is strip1 and the blue box is the strip 2. Left: Inner Mongolia. Right: Yellow River.

TABLE II
STRIP ADJUSTMENT RESULTS FOR THE INNER MONGOLIA DATASET

Methods		Original	Point-to-point ICP	Point-to-plane ICP	FPFH	Our
Distance	Mean(m)	0.219	0.106	0.100	0.244	**0.081**
	RMSE(m)	0.280	0.135	0.144	0.281	**0.093**
DSM	Mean(m)	0.412	0.171	0.173	11.449	**0.164**
	RMSE(m)	0.749	0.517	0.518	13.799	**0.515**

TABLE III
STRIP ADJUSTMENT RESULTS FOR THE YELLOW RIVER DATASET

Methods		Original	Point-to-point ICP	Point-to-plane ICP	FPFH	Our
Distance	Mean(m)	0.191	**0.093**	0.097	0.260	0.094
	RMSE(m)	0.224	**0.114**	0.119	0.292	0.114
DSM	Mean(m)	0.280	0.069	0.065	3.455	**0.059**
	RMSE(m)	0.342	0.128	0.124	4.288	**0.121**

and DSM discrepancies for the data treated with the proposed method are smaller than the results of the other methods. In the Yellow River data, the mean of Euclidean distance is reduced from 0.191 to 0.094 m, and the mean of DSM discrepancies is reduced from 0.280 to 0.059 m. Our method outperforms the comparison methods in terms of the DSM discrepancies, while opposite results are shown in terms of the Euclidean distance metric. The reasons for the above phenomenon are as follows. The ICP-based method is achieved by minimizing the Euclidean distance for the point cloud transformation, whereas our method is a further spatial alignment on top of ICP. Because of the large gap between the number of ground points and nonground points, the ground points make a greater contribution to the calculation of the point cloud transformation matrix. This phenomenon leads to the fact that the spatial alignment is not optimal even if the mathematical calculation is optimal. In fact, spatial alignment is more important than the minimization of Euclidean distance for strip adjustment. The improvement in DSM discrepancies, although reduced by only about 1 cm, is important for the improvement in strip adjustment. As shown in Figs. 5 and 6, although the ICP-based method is not spatially aligned, there is still a clear majority of points that are close to overlap, especially for the ground points. The ground points occupy a large proportion of the overall data, resulting in an insignificant improvement in the quantitative metrics. The FPFH method cannot reduce the difference between the strips and even makes the results worse in both datasets. This is because FPFH relies on the construction of LRF, which is unstable for such nonurban point clouds.

2) Qualitative Evaluation: Figs. 5 and 6 illustrate the experimental results of the two datasets before and after strip adjustment visually. It can be seen that there are obvious spatial discrepancies between the two strips in the original data. In particular, the points from different point clouds are divided into two layers. After strip adjustment, spatial discrepancies between the two strips are reduced obviously. By comparing the local details, the results obtained by the

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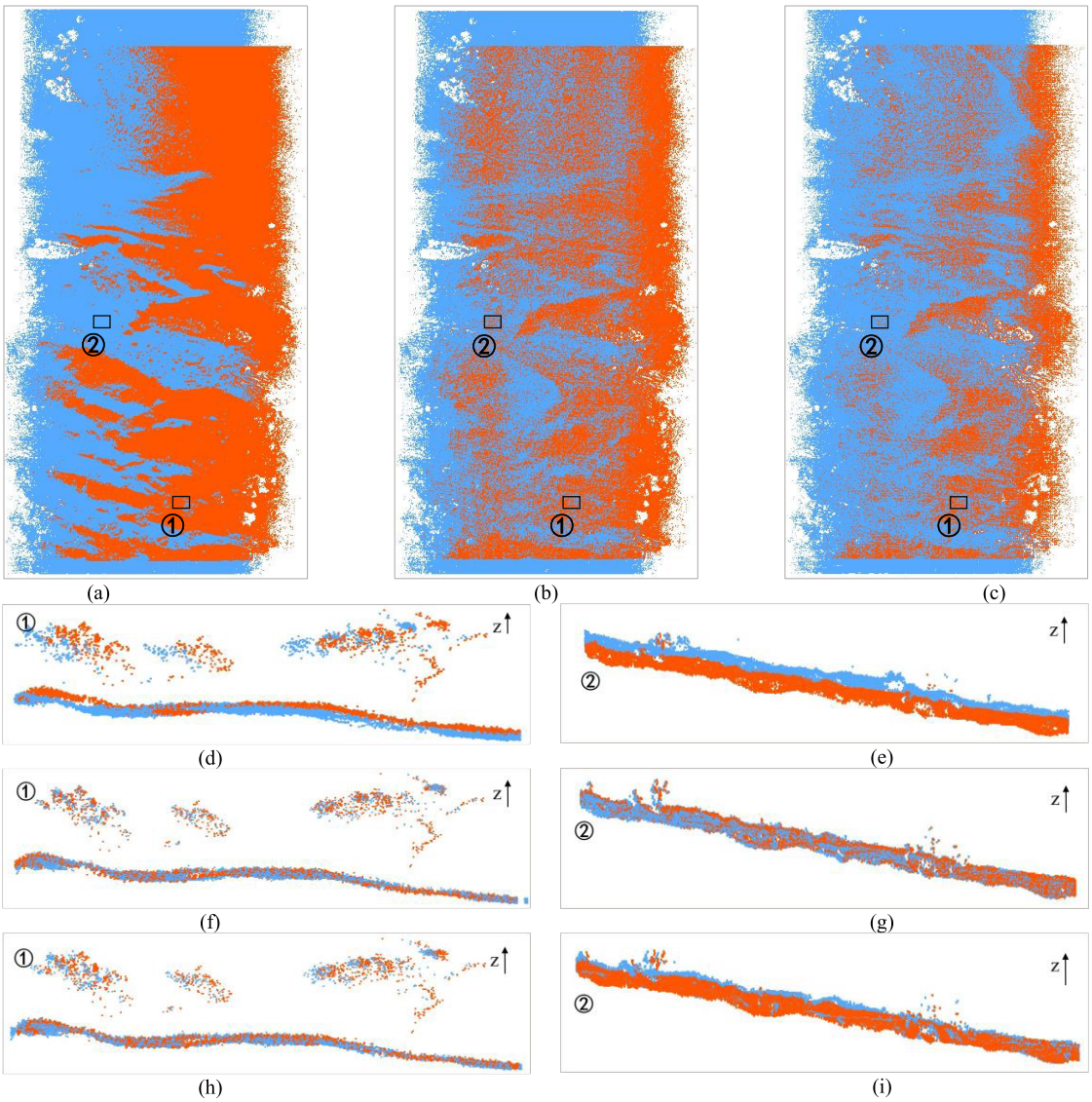


Fig. 5. Strip adjustment result comparison for the Inner Mongolia dataset. The orange points represent P_1, and the blue points represent P_2. (a) Original data. (b) Proposed method. (c) Point-to-point ICP. (d) and (e) Local areas of original data. (f) and (g) Local areas of the proposed method. (h) and (i) Local areas of point-to-point ICP.

proposed method are more accurately aligned in space than those obtained by the point-to-point ICP.

Fig. 7 illustrates the discrepancies between the DSMs of overlapping by two strips in the two datasets. The results of Inner Mongolia show that some parts of spatial discrepancies are eliminated by strip adjustment. Significant differences, however, also exist in slope terrain after the process by point-to-point ICP. Even in gently rolling topography, obvious spatial discrepancies still exist, which mainly focus on the nonground objects. The proposed method performs excellently compared with the original data and the results of point-to-point ICP, both in terrain and nonground objects. In Yellow River, although the point-to-point ICP reduces certain terrain discrepancies, the remaining spatial discrepancies still exist and mainly focus on the ground objects' edge, and remaining spatial discrepancies are further reduced by our method. It can be concluded that the proposed method effectively reduces the discrepancies between strips. This is essential for high-quality large-area DSM generation when multiple strips are required.

C. Descriptiveness and Robustness Evaluation

In this part, the influence of the maximum shell range $r_{\max}$, shell thickness t, and the point density on the descriptiveness of the point feature descriptors are analyzed. In addition, the robustness of $r_{\max}$, t, and the overlapping ratio of adjacent strips on strip adjustment are also tested.

1) SSPF Descriptiveness Analysis: To analyze the descriptiveness of the SSPF, we used the Recall versus 1-Precision curve (RP Curve) for different parameter setting conditions. First, key points in P_1 and their closest corresponding points in adjusted P_2 are selected, and they are represented by corresponding key points. Second, the SSPFs are calculated using different parameters, and each SSPF of key points is matched against all SSPFs of corresponding points to find

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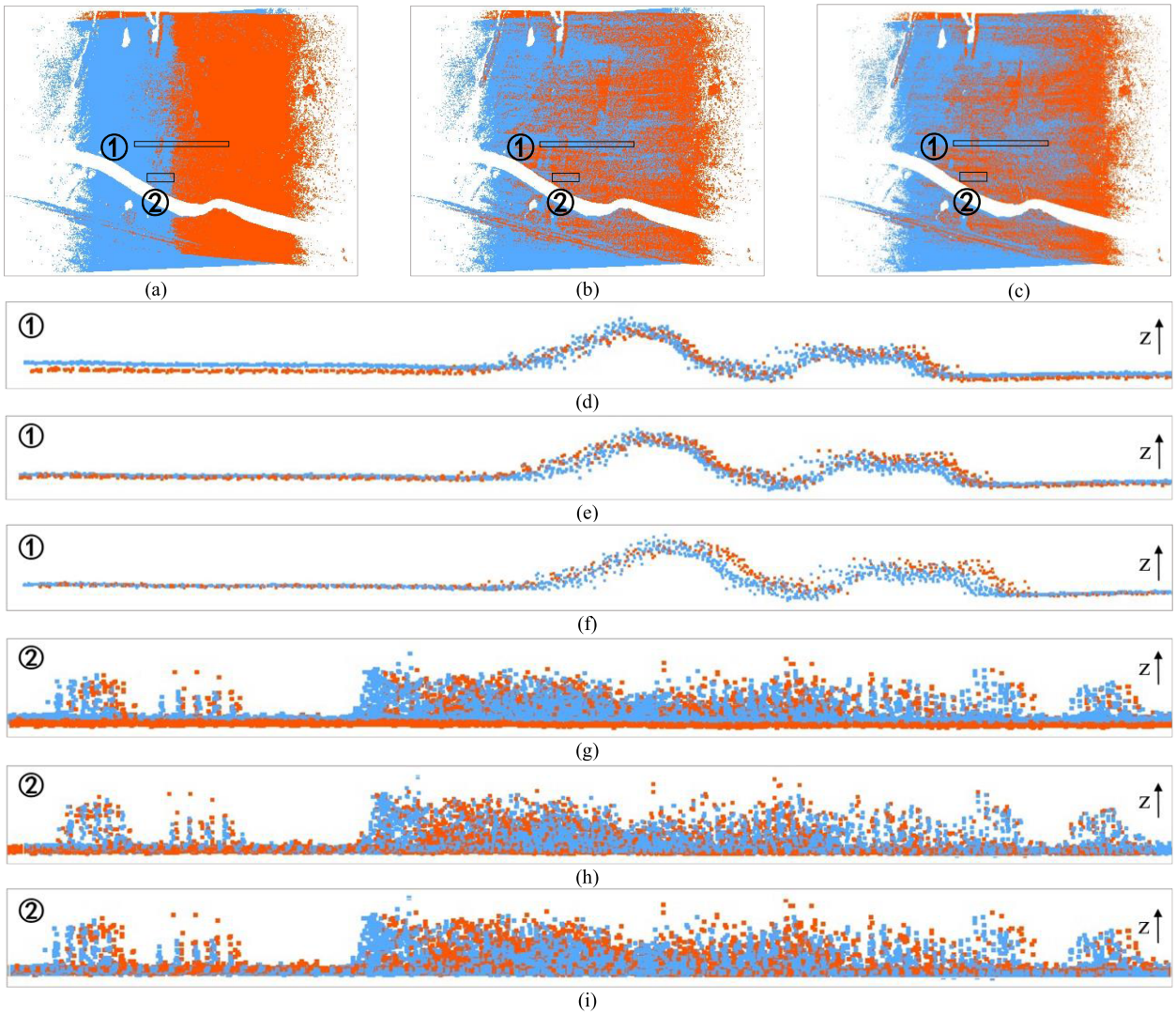


Fig. 6. Strip adjustment result comparison for the Yellow River dataset. The orange points represent P_1, and the blue points represent P_2. (a) Original data. (b) Proposed method. (c) Point-to-point ICP. (d) and (g) Local areas of original data. (e) and (h) Local areas of the proposed method. (f) and (i) Local areas of point-to-point ICP.

the most similar correspondences. If the distance between two SSPFs is smaller than a threshold τ, the two points are considered a feature match. Finally, the feature match is defined as a correct feature match if the two points are corresponding key-points. The precision is calculated as the ratio of the number of correct feature matches to the total number of feature matches. The recall is calculated as the ratio of the number of correct feature matches to the number of corresponding key points. By varying the threshold τ, the RP Curve can be generated.

The RP curves of the maximum shell range $r_{\max}$ and shell thickness t are shown in Fig. 8(a)–(h). In terms of the maximum shell range $r_{\max}$, as it increases, the recall and precision become higher, and the corresponding SSPF becomes more descriptive on both two datasets. This is because more points within a larger range of the neighborhood are used as $r_{\max}$ increases, which makes the dimensionality of SSPF higher; however, when $r_{\max} = 5$ m, the descriptiveness of SSPF becomes lower, which is the result of the limited spatial information available. In terms of the spherical shell thickness t, high descriptiveness is obtained at $t = 1$ m. As t continues to increase to 2.5–5 m, the descriptiveness is still high for a large maximum range, but it becomes lower when $r_{\max}$ is smaller. In general, the descriptiveness becomes higher when the $r_{\max}$ is larger, but this also brings more computational effort. Setting $r_{\max}$ to around 10 m and t to around 1 m allows for a balance between computation and descriptiveness. Note that such parameter settings are consistent in both data.

Fig. 9 illustrates the RP curves of the point density. Down-sampling to 20, 15, 10, and 5 pts/m^2 was applied for both two datasets, then let $r_{\max} = 10$ m and $t = 1$ m. By comparing recalls of the two datasets, it can be observed that the SSPF is more descriptive with higher point density. In Inner Mongolia, the point density of the original data is about 245 pts/m^2, which is much higher than the down-sampled data, but the SSPFs calculated with the data of 20 and 15 pts/m^2 still gain good descriptiveness, which is slightly worse than the original data. In Yellow River, the descriptiveness of the down-sampled

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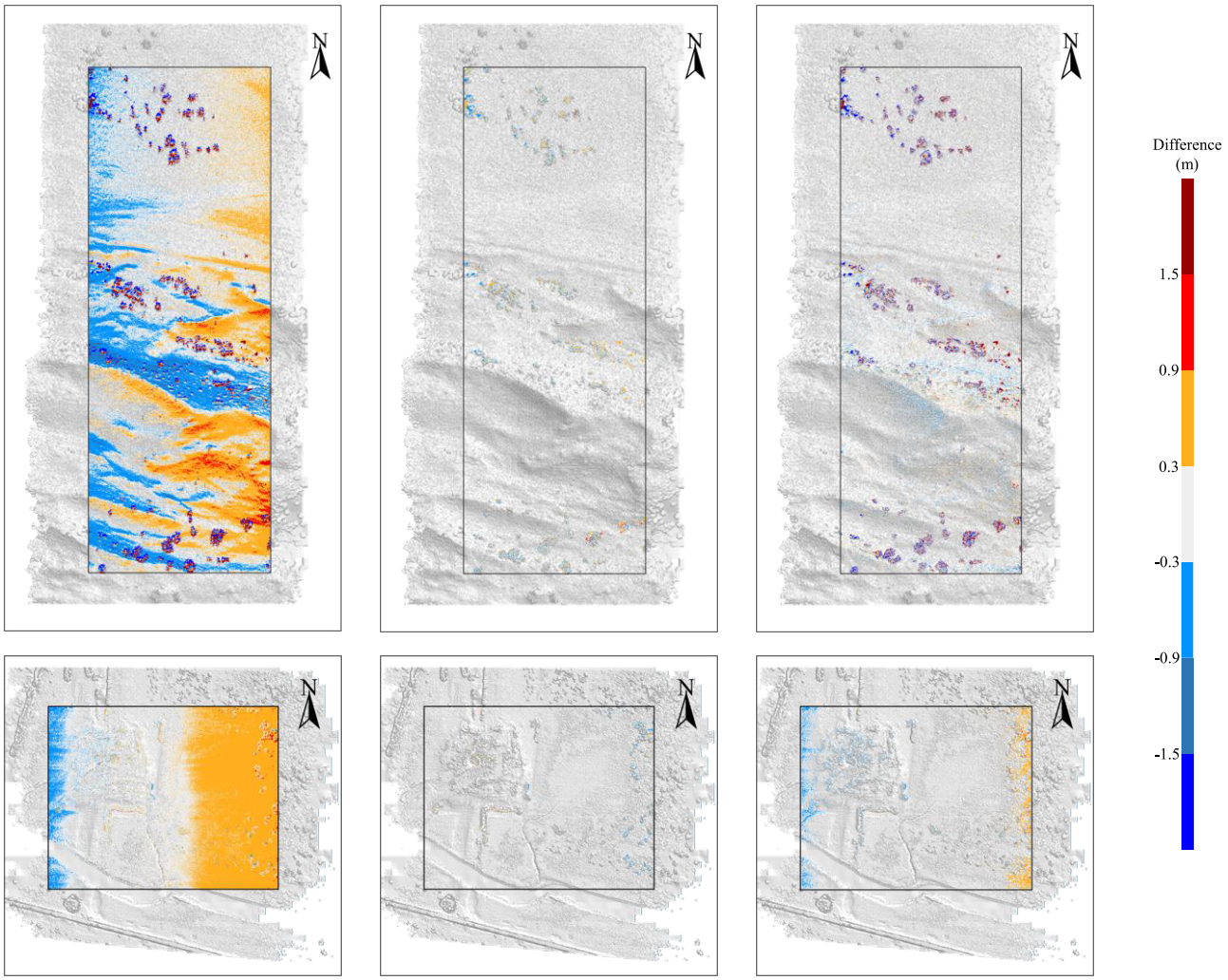


Fig. 7. DSM discrepancies of the sharing overlapping between two strips, the inner rectangle is the extent of the overlapping area. (Top) Inner Mongolia. (Bottom) Yellow River. From left to right: original, the proposed method, and point-to-point ICP.

data with 20 and 15 pts/m^2 density is almost equal to the original data (21 pts/m^2). Although the descriptiveness of SSPF calculated by lower density datasets (10 and 5 pts/m^2) becomes poor, the point clouds obtained by almost all UAV LiDARs are more than needed.

2) Strip Adjustment Robustness Test: To test the robustness of the proposed method, different maximum shell range $r_{\max}$ and spherical shell thickness t are set to adjust strips, and lower overlapping ratio point clouds (the range of key-points selection is limited in the original data) are used to simulate various realistic conditions. The quantitative metric is the mean of the DSM discrepancies, which provides a better evaluation of spatial alignment. Fig. 10 illustrates the impact of $r_{\max}$ and t on adjustment results. According to the results of the two datasets, the mean value of DSM discrepancies decreases as $r_{\max}$ increases in an overall trend. When $r_{\max} =$ 5 m, the strip adjustment results were worse than those obtained by the ICP-based method in both datasets. This is because the maximum shell range of the neighborhood limits the scale and dimensionality of the features, making the corresponding points unsuccessfully paired, and thus, leading to poorer results. At a maximum shell range of 10 m, our method achieves an improvement compared to other methods. As for the thickness of the shell, good and stable results were obtained when the thickness of the shell was $t = 1$ m and $t = 2.5$ m. Whereas when $t = 0.5$ m, the shell contains too few points, and at $t = 5$ m, the thicker shell results in few feature dimensions and hiding spatial features, which makes the results unsatisfactory.

TABLE IV
DSM DISCREPANCIES ABOUT THE OVERLAPPING RATIO

Overlapping ratio		50%	40%	30%	20%
Inner Mongolia	Mean(m)	0.167	0.165	0.174	0.184
	RMSE(m)	0.514	0.515	0.518	0.573
Yellow River	Mean(m)	0.060	0.064	0.065	0.072
	RMSE(m)	0.120	0.122	0.125	0.133

We also analyzed the influence of different overlapping ratios on the performance of strip adjustment. The results in Table IV show that the DSM discrepancies rise slightly when the overlapping ratios drastically increase. In particular, our method still yields favorable results for the data of the

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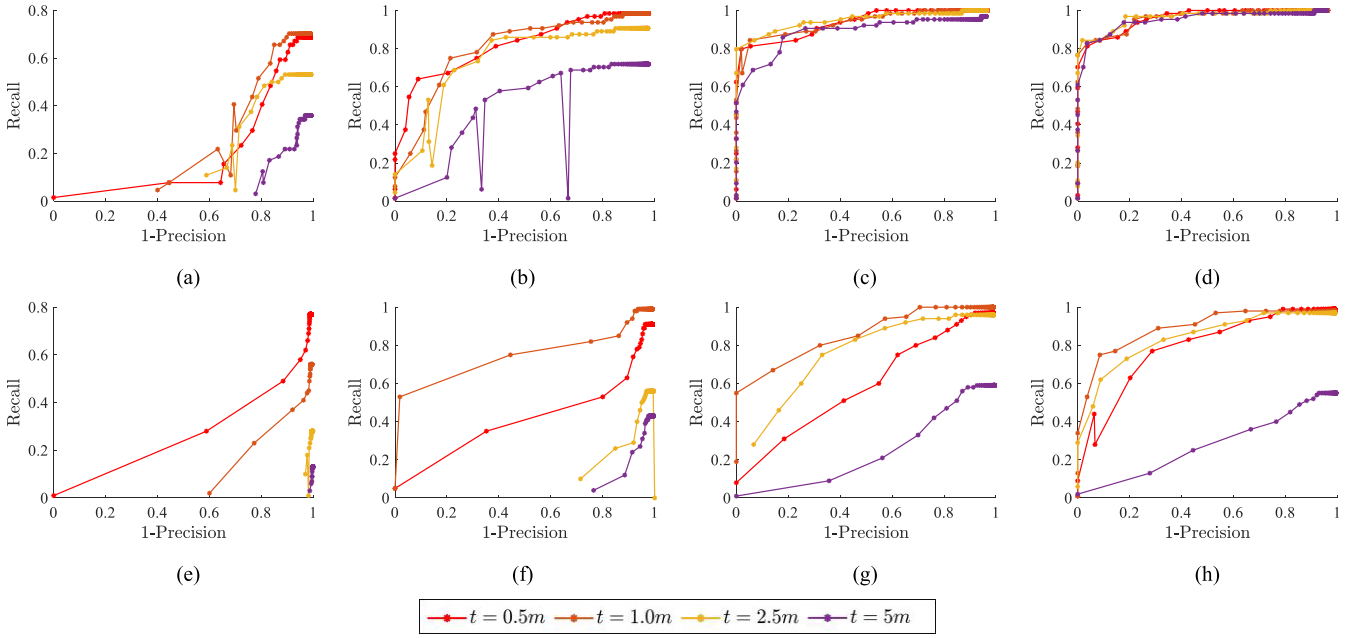


Fig. 8. RP curves of the maximum shell range and shell thickness. (Top) Inner Mongolia. (Bottom) Yellow River. (a) $r_{\max}$ =5 m. (b) $r_{\max}$ =10 m. (c) $r_{\max}$ =15 m. (d) $r_{\max}$ =20 m. (e) $r_{\max}$ =5 m. (f) $r_{\max}$=10 m. (g) $r_{\max}$ =15 m. (h) $r_{\max}$ =20 m.

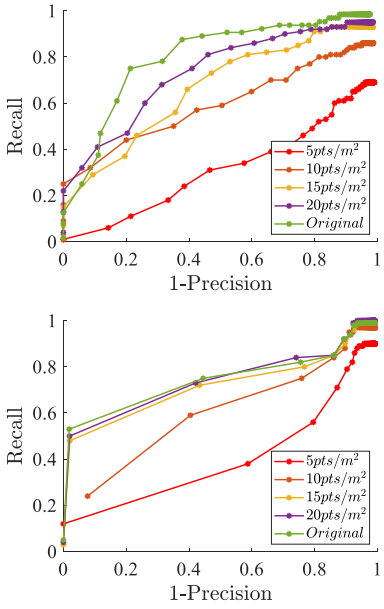


Fig. 9. RP curves of point density. (Top) Inner Mongolia. (Bottom) Yellow River.

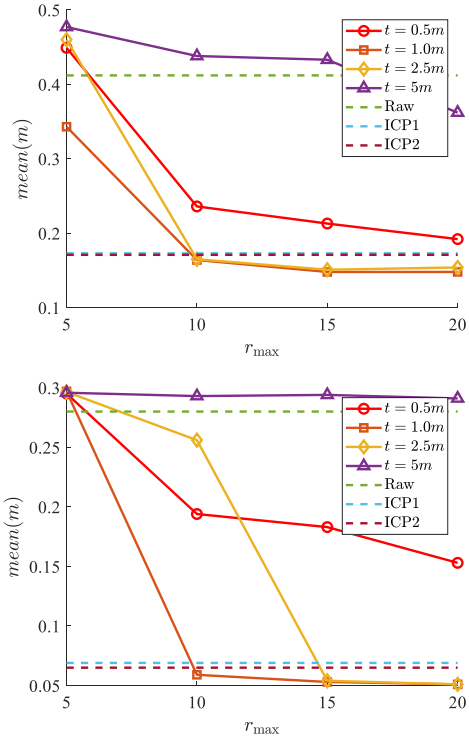


Fig. 10. Mean of DSM discrepancies with increasing $r_{\max}$ and different t. (Top) Inner Mongolia. (Bottom) Yellow River.

20% ratio (only 0.020 and 0.013 m increase compared to the original data), which verifies the effectiveness of the proposed method under low overlapping ratio data.

3) Four SSPF Property Analysis: To verify the properties of SSPF, we simulate four interfered point clouds based on P_1: 1) performing random rotations, 2) uniformly sampling, 3) randomly sampling, and 4) manually adding noise points. The SSPF is calculated with each of the interferences, then the SSPFs of P_1 and four simulation datasets are compared. The parameters for SSPF calculation are set as $r_{\max} = 10$ m and $t = 1$ m [i.e., $\mathbf{f}_1 \in \mathbb{R}^{20\times 1}$ in (2)], and the parameters for four simulation datasets are set as follows: 1) Random rotation of 0°–10° around x, y, and z axis, respectively. 2) Down-sampling the point cloud with a distance threshold of 0.4 m (compared to 0.2 m at the first time). 3) Random sampling with a rate of 60%. 4) Adding 0.1% noise points randomly distributed around the main point cloud (considering the outlier removal). The evaluation is as follows: we first define that the features in the SSPF with a difference of less than 0.05 m are considered to remain constant. The number of

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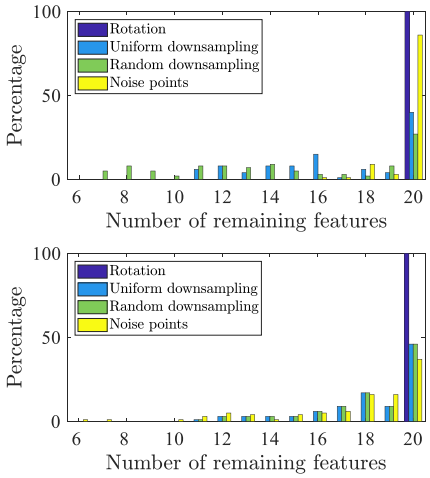


Fig. 11. Histogram of the number of remaining features in four simulation point clouds. (Top) Inner Mongolia. (Bottom) Yellow River.

features remaining constant in SSPF varies between 0 and 20. We use the histogram to represent the four properties of SSPF. Fig. 11 illustrates the histograms of the number of remaining features in four simulation point clouds. It can be seen that in both datasets, the key points maintaining more than 15 features occupy a huge proportion, with 100%, 74%, 86%, and 100% corresponding to four simulations, respectively, of the Inner Mongolia dataset and 100%, 90%, 90%, and 84% of the Yellow River dataset. In particular, all the 20 dimensions of SSPF of the rotated simulation dataset remained unchanged, which is consistent with the SSPF extraction method. The above analysis demonstrates the four important properties of SSPF.

IV. Conclusion

In this article, a new strip adjustment method for nonurban scenes is proposed to reduce the spatial discrepancies between strips. To overcome the difficulty of aligning nonurban point clouds, a novel point feature descriptor called "SSPF" is proposed, and it is combined with the point cloud coordinates to generate embedded features, which are further used in two-stage matching. Instead of constructing LRF, which is unreliable in nonurban scenes, the SSPF describes points by dividing the neighborhood of the points into spherical shells of different scales and calculating distances that represent spatial information. Our method solves the problem of difficult spatial alignment when the number of ground points and nonground points differs significantly. Validated on two challenging datasets of different nonurban scenes acquired by different UAV LiDAR systems, the proposed method reduces the DSM discrepancies by 0.252 and 0.221 m from the original data, respectively, and its effectiveness and superiority are proved compared to other methods. Our method also has one limitation: For terrestrial LiDAR point clouds, the occlusion makes ground objects' profiles incomplete, thereby reducing the descriptiveness of the proposed SSPF.

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Yanfeng Gu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined as a Lecturer with the School of Electronics and Information Engineering, HIT. He was appointed as an Associate Professor at the same institute in 2006; meanwhile, he was enrolled in the First Outstanding Young Teacher Training Program of HIT. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California, Berkeley, CA, USA. He is currently a Professor with the Department of Information Engineering, HIT. He has published more than 100 peer-reviewed articles, four book chapters, and he is the inventor or coinventor of 20 patents. His research interests include space intelligence remote sensing and information processing, multimodal hyperspectral remote sensing, and spaceborne time-series image processing.



Zhen Xiao received the B.E. degree in electronics and information engineering from the Harbin Institute of Technology, Harbin, China, in 2020, where he is currently pursuing the Ph.D. degree in information and communication engineering.

His research interests include UAV LiDAR point cloud processing and the application of point cloud in remote sensing.



Xian Li (Member, IEEE) received the Ph.D. degree in instrument science and technology from the Harbin Institute of Technology (HIT), Harbin, China, in 2021.

From 2018 to 2020, he was a Doctoral Researcher with the Department of Telecommunications and Information Processing, Ghent University, Ghent, Belgium, supported by the China Scholarship Council. He is currently an Assistant Professor with the School of Electronics and Information Engineering, HIT. His research interests include deep learning, hyperspectral remote sensing, and image analysis.

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